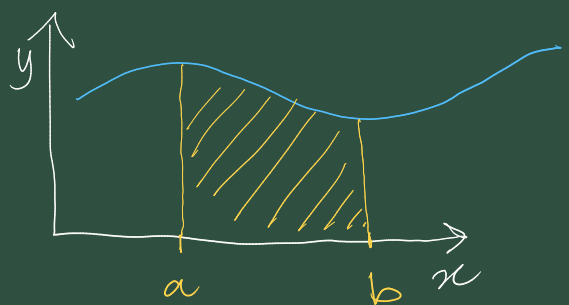
Multiple integrals (Double integrals)

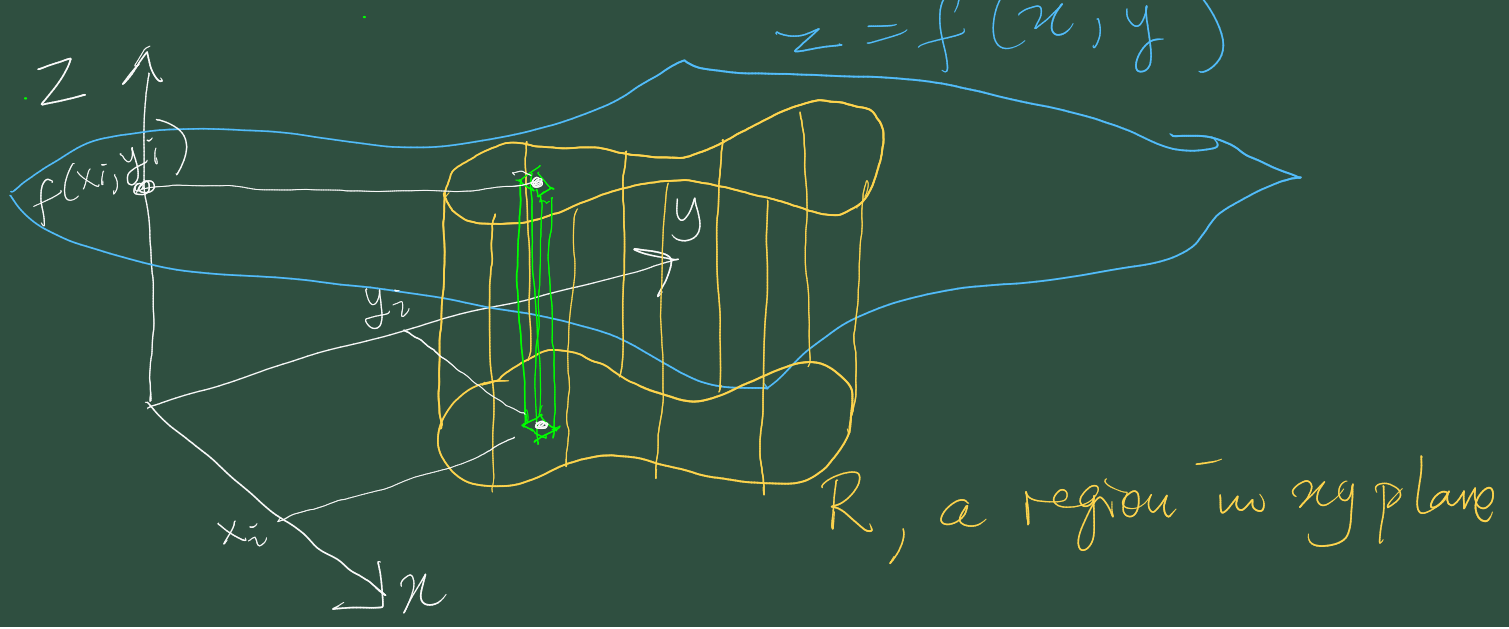
Double integrals generalise to two-variables the concept of definite integral of a single variable function.



$\int_a^b f(x)\,dx$ gives the area 'underneath the curve'.

Let's consider a function $f(x, y)$, and the surface in 3-dimensions defined by

$$z = f(x, y)$$

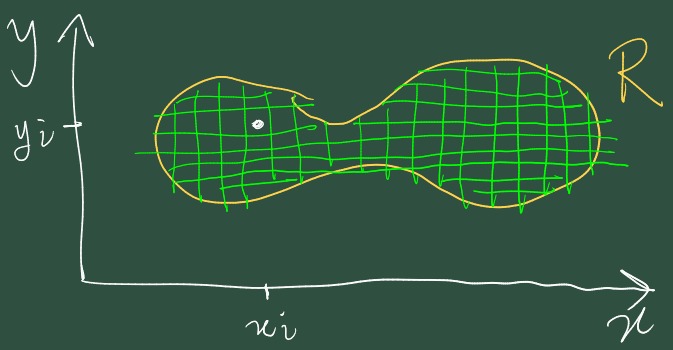


The volume V between the surface and xy-plane over the region R will be represented by the 'double integral'

$$V = \iint_R f(x,y)\, dx\,dy$$

This can be formally defined in a similar way to Riemann integration in single-variable case using a partition of R into rectangles τ_i so that

$$R \approx \bigcup_i \tau_i$$



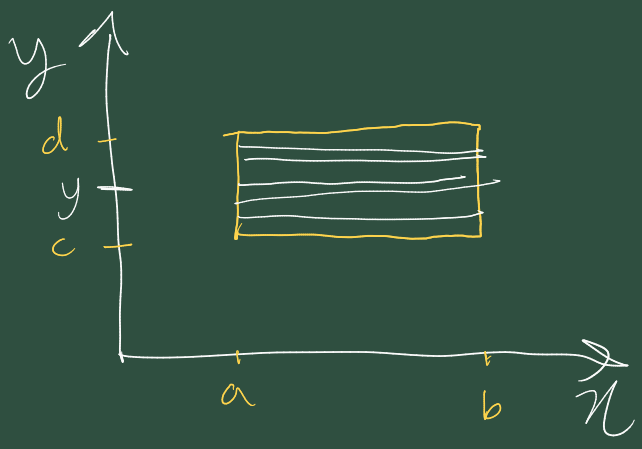
In each τ_i select a representative point (x_i, y_i). Over each τ_i position a rectangular prism of height $f(x_i, y_i)$ and define the integral as

$$V = \lim_{\substack{\text{size } \tau_i \to 0 \\ \text{and } \#\tau_i \to \infty}} \sum_i (\text{area } \tau_i)\, f(x_i, y_i)$$

We won't actually calculate such integrals using the limit process directly, rather we

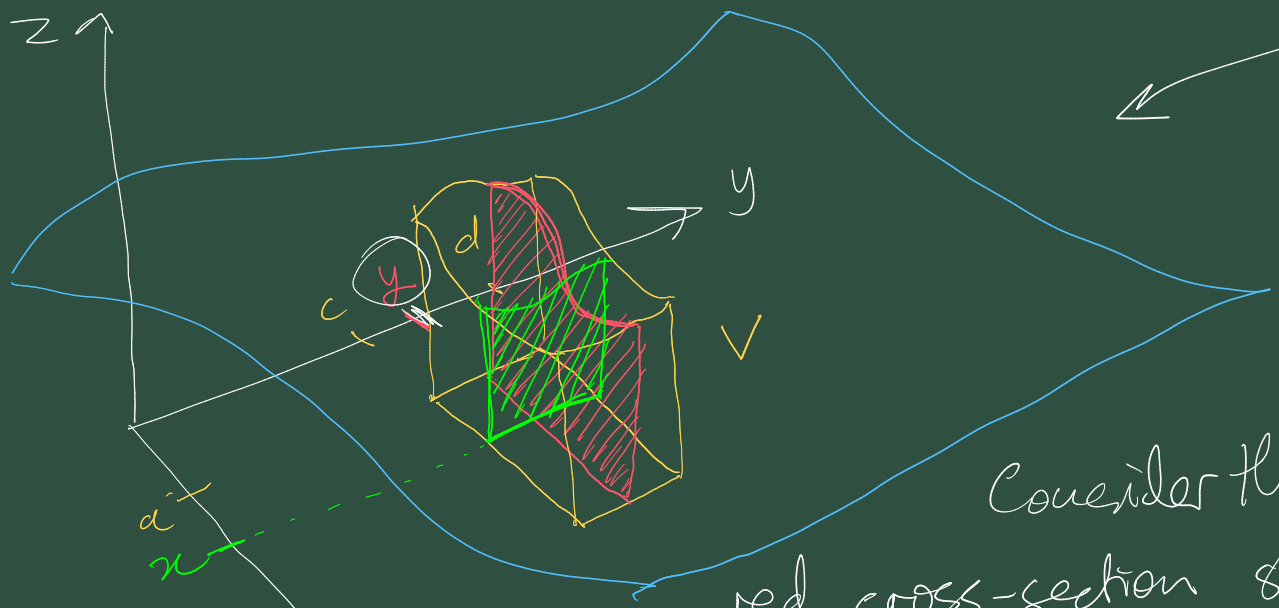
will use 'repeated (single-variable) integration' to calculate them.

Simplest case is when R is itself a rectangle, aligned with x, y axes.



fixed location of the green curtain

$$\int_c^d f(x,y)\,dy$$



Consider the red cross-section shown which is located at a particular y value satisfying $c \le y \le d$

This cross-section has area (the location of the cross-section)

$$\int_a^b f(x,y)\,dx$$

Idea is to express the volume, over R, as the

continuous sum of such areas / cross-sections taken at y-values from c, ..., to d

ie. $$V = \int_c^d \left(\int_a^b f(x,y)\,dx \right) dy$$

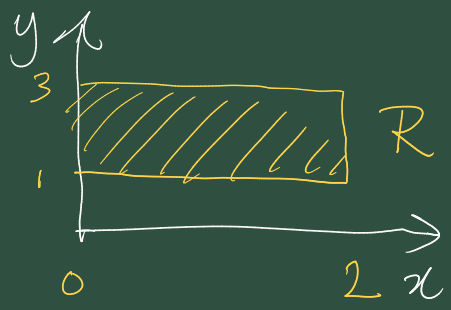
$\int_a^b f(x,y)\,dx$: cross-sectional area.

There are typically evaluated by performing the inner integral first, and then the outer integral.

Example.

Consider $f(x,y) = 2xy + 4x + 3y + 1$

Integrate this over the rectangular region R defined by $0 \le x \le 2$ and $1 \le y \le 3$.



$$V = \iint_R f(x,y)\,dx\,dy$$

$$= \int_1^3 \left(\int_0^2 (2xy + 4x + 3y + 1)\,dx \right) dy$$

$$= \int_1^3 \left(\left[x^2 y + 2x^2 + 3xy + x \right]_{x=0}^{x=2} \right) dy$$

$$= \int_1^3 (4y + 8 + 6y + 2)\,dy$$

$$= \int_1^3 (10y + 10)\, dy$$

$$= \left[5y^2 + 10y \right]_1^3$$

$$= 45 + 30 - (5 + 10)$$

$$= 60$$

Note, we could have expressed this double integral as the repeated integral in the other order

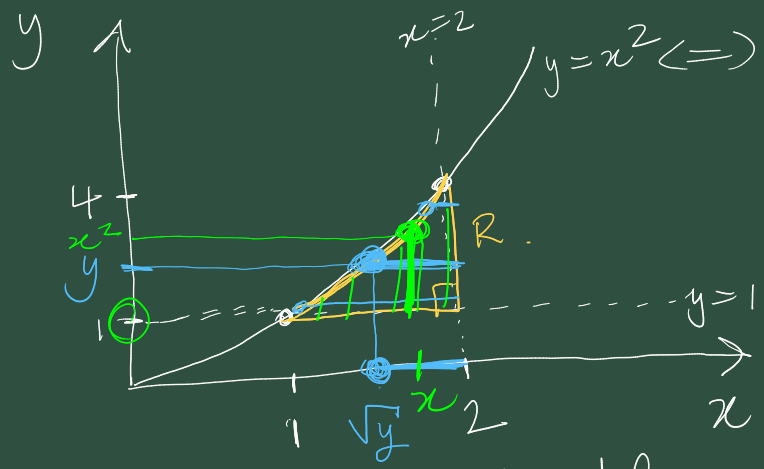
$$V = \iint_R f(x, y)\, dy\, dx$$

$$= \int_a^b \left(\int_c^d f(x, y)\, dy \right) dx$$

Can check that this will evaluate to 60 also.

Non-rectangular region, typically described by a collection of boundary curves.

Example Integrate $f(x, y) = 4x^3 + 4y^3$ over R, the finite region bounded by the curves $x = 2$, $y = 1$ and $y = x^2$



Lecture at 12:00

We want the double integral V

$$V = \iint_R f(x,y)\,dx\,dy$$

$$= \int_1^4 \left(\int_{\sqrt{y}}^{2} (4x^3 + 4y^3)\,dx \right) dy$$

(once we have the limits set-up correctly the ~~calculation~~ evaluation should proceed as normal)

$$= \int_1^4 \left(\left[x^4 + 4xy^3 \right]_{x=\sqrt{y}}^{x=2} \right) dy$$

$$= \int_1^4 \left((16 + 8y^3) - (y^2 + 4y^{7/2}) \right) dy$$

$$= \left[16y + 2y^4 - \frac{y^3}{3} - 4\frac{2}{9}y^{9/2} \right]_1^4$$

$$= 64 + 512 - \frac{64}{3} - \frac{8}{9}2^9 - \left(16 + 2 - \tfrac{1}{3} - 8/9\right)$$

$= \frac{745}{9}$

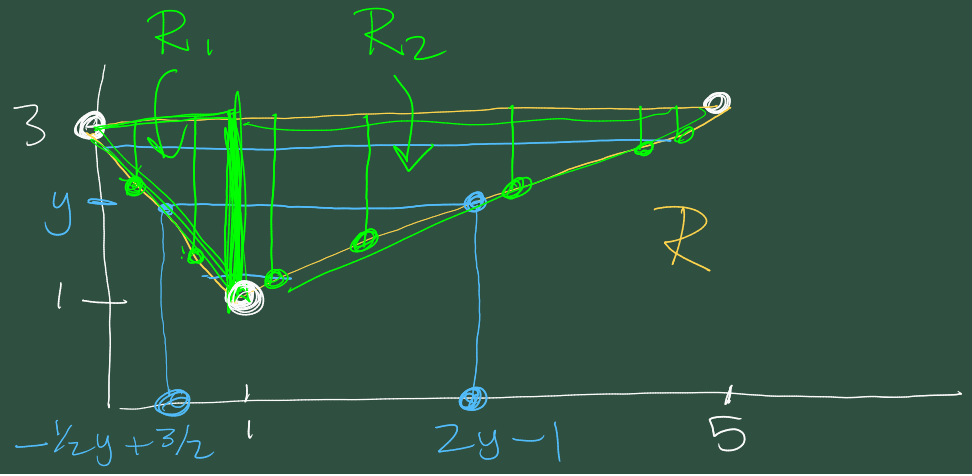
What does the repeated integral in the other order look like?

$$V = \int_1^2 \left(\int_1^{x^2} (4x^3 + 4y^3)\, dy \right) dx$$

$$= \int_1^2 \left(\int_1^{x^2} (4x^3 + 4y^3)\, dy \right) dx$$

Example non-rectangular and shows the need to sometimes decompose the region

Consider integrating a function $f(x,y)$ over the triangular region R with vertices at $(1,1)$, $(5,3)$, $(0,3)$



We'll need the boundary curves.

$(0,3) \to (5,3)$: $y = 3$

$(0,3) \to (1,1)$: $y = -2x + 3 \iff x = -\frac{1}{2}y + \frac{3}{2}$

$(1,1) \to (5,3)$: $y = \frac{1}{2}x + \frac{1}{2} \iff x = 2y - 1$

$$V = \iint_R f(x,y)\, dx\, dy$$

$$= \int_1^3 \left(\int_{-\frac{1}{2}y + \frac{3}{2}}^{2y-1} f(x,y)\, dx \right) dy$$

and

$$V = \iint f(x,y)\, dy\, dx$$

$$= \int_0^5 \left(\int_{?}^{3} f(x,y)\, dy \right) dx$$

No single expression that can go here

In order to achieve this we have to express the double integral as:

$$V = \iint_R f(x,y)\, dy\, dx$$

$$= \iint_{R_1} f(x,y)\,dy\,dx + \iint_{R_2} f(x,y)\,dy\,dx$$

where $R = R_1 \cup R_2$ is the disjoint giving R as the union of the two triangular parts separated by $x=1$

$$= \int_0^1 \left(\int_{-2x+3}^{3} f(x,y)\,dy \right) dx$$

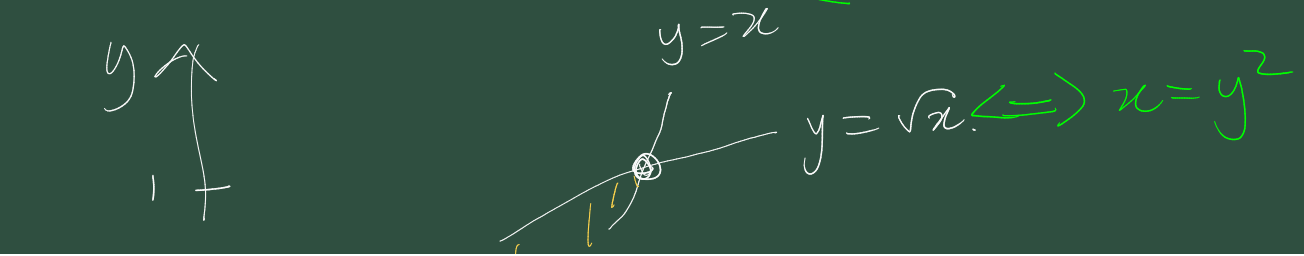
$$+ \int_1^5 \left(\int_{\frac{1}{2}x+\frac{1}{2}}^{3} f(x,y)\,dy \right) dx.$$

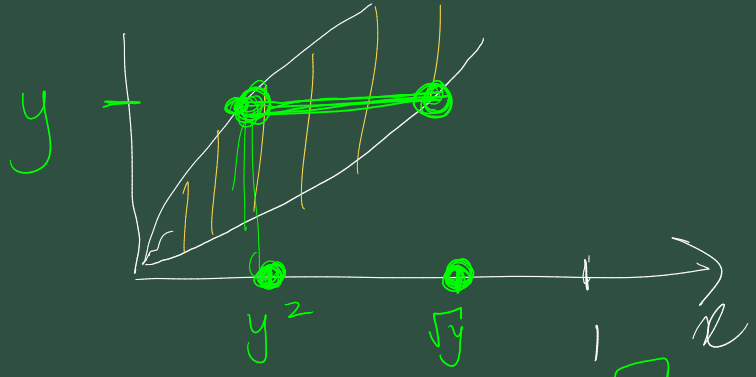
Q3 (c). $V = \iint_R xy^2\,dx\,dy$

where R is the finite region in the first quadrant bounded by $y=x^2$ and $x=y^2 \iff y=\sqrt{x}$

$y=x^2 \iff x=\sqrt{y}$

$y=\sqrt{x} \iff x=y^2$





$$V = \int_0^1 \left(\int_{y^2}^{\sqrt{y}} xy^2 \, dx \right) dy$$

$$= \int_0^1 \left[\frac{x^2 y^2}{2} \right]_{x=y^2}^{x=\sqrt{y}} dy$$

$$= \int_0^1 \frac{1}{2} (y^3 - y^6) \, dy$$

$$= \frac{1}{2} \left(\frac{y^4}{4} - \frac{y^7}{7} \right)_0^1 = \frac{1}{2} \left(\frac{1}{4} - \frac{1}{7} \right)$$

$$= \frac{1}{2} \left(\frac{7}{28} - \frac{4}{28} \right) = \frac{1}{2} \left(\frac{3}{28} \right)$$

$$= \frac{3}{56}$$